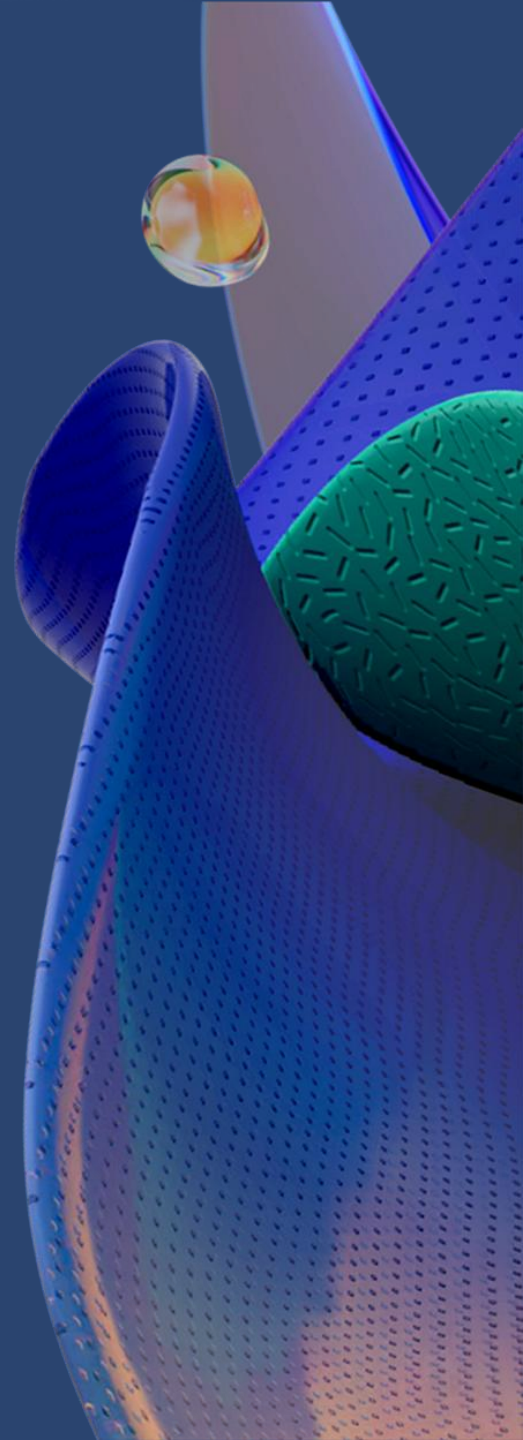


Secure data end-to-end with Microsoft Fabric and OneLake



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Microsoft

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Did Microsoft abandon OneSecurity?

I don't see it on any diagrams (even new diagrams from FabCon), any docs, and not a word about it across any of the announcement blogs today. Does this mean that data security will just continue to be managed and enforced differently for each engine (Power BI, Spark, Warehouse, KQL, etc.) ? If so, this is honestly pretty frustrating as one of the best parts of the original Fabric vision was "secure once".

Agenda

- 1 OneLake and security in Microsoft Fabric
- 2 Security model details
- 3 API details
- 4 Shortcuts and integration patterns
- 5 Managing OneLake security
- 6 Wrap up

OneLake and security in Microsoft Fabric

OneLake for all data

"The OneDrive for data"



OneDrive
for documents

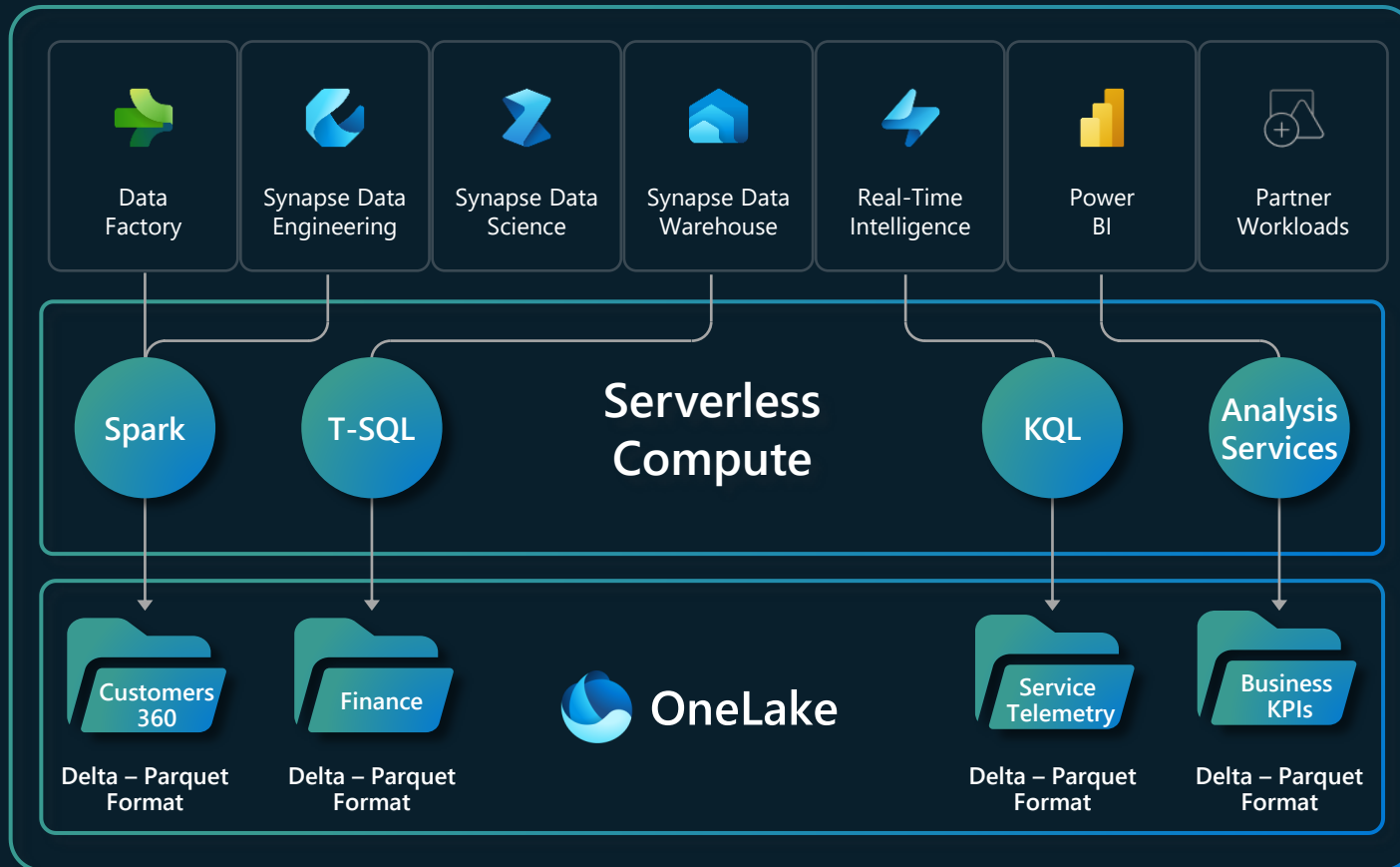


OneLake
for data

OneLake provides a data lake as a service
without you needing to build it

One Copy for all computes

Real separation of compute and storage



All the compute engines store their data automatically in OneLake as data items.

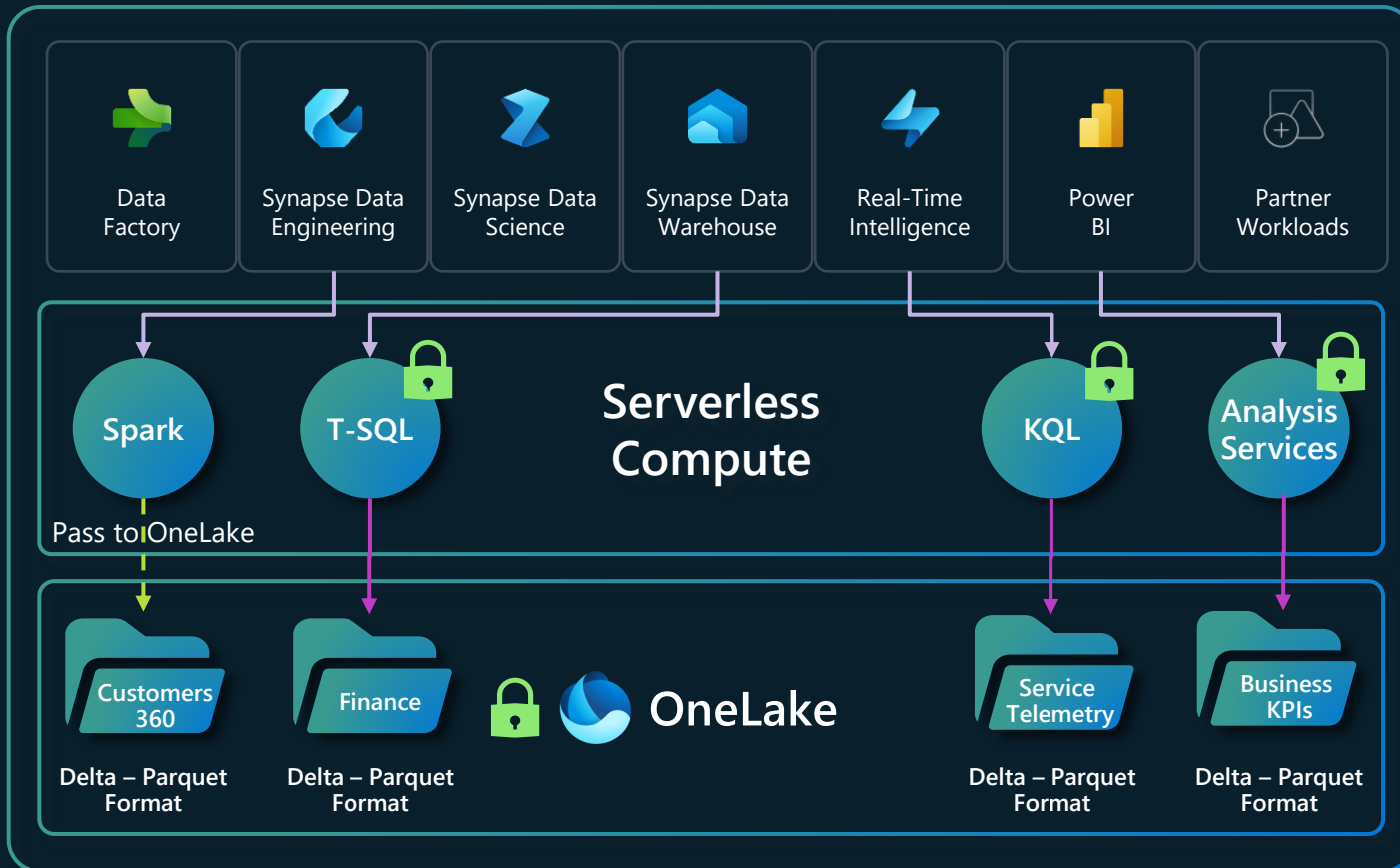
The data is stored in a single common format.

Delta – Parquet, an open standards format, and it is the storage format for all tabular data in Fabric.

All the compute engines have been fully optimized to work with Delta Parquet as their native format.

One Copy, many security definitions

Enabling fine-grained security at each engine



Each engine in Fabric provides its own robust set of security capabilities.

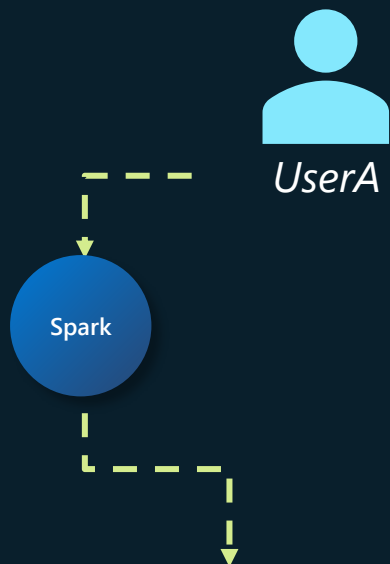
OneLake also brings data lake security, applicable to engines like Spark.

Each Fabric engine relies on one of two modes when reading data from OneLake.

One Copy, many security definitions

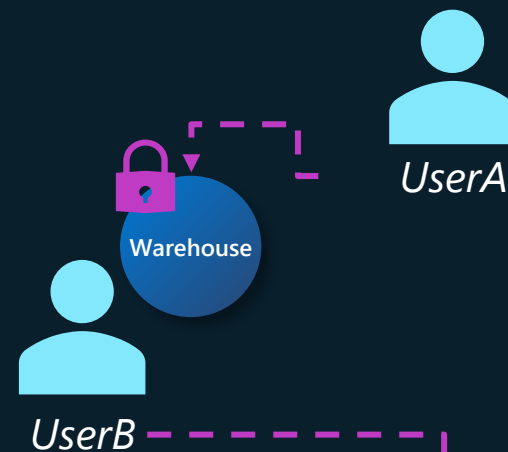
Enabling fine-grained security at each engine

Passthrough



In passthrough mode, the identity of the querying user is used to evaluate the access in OneLake. In this mode, OneLake security fully determines the user's access.

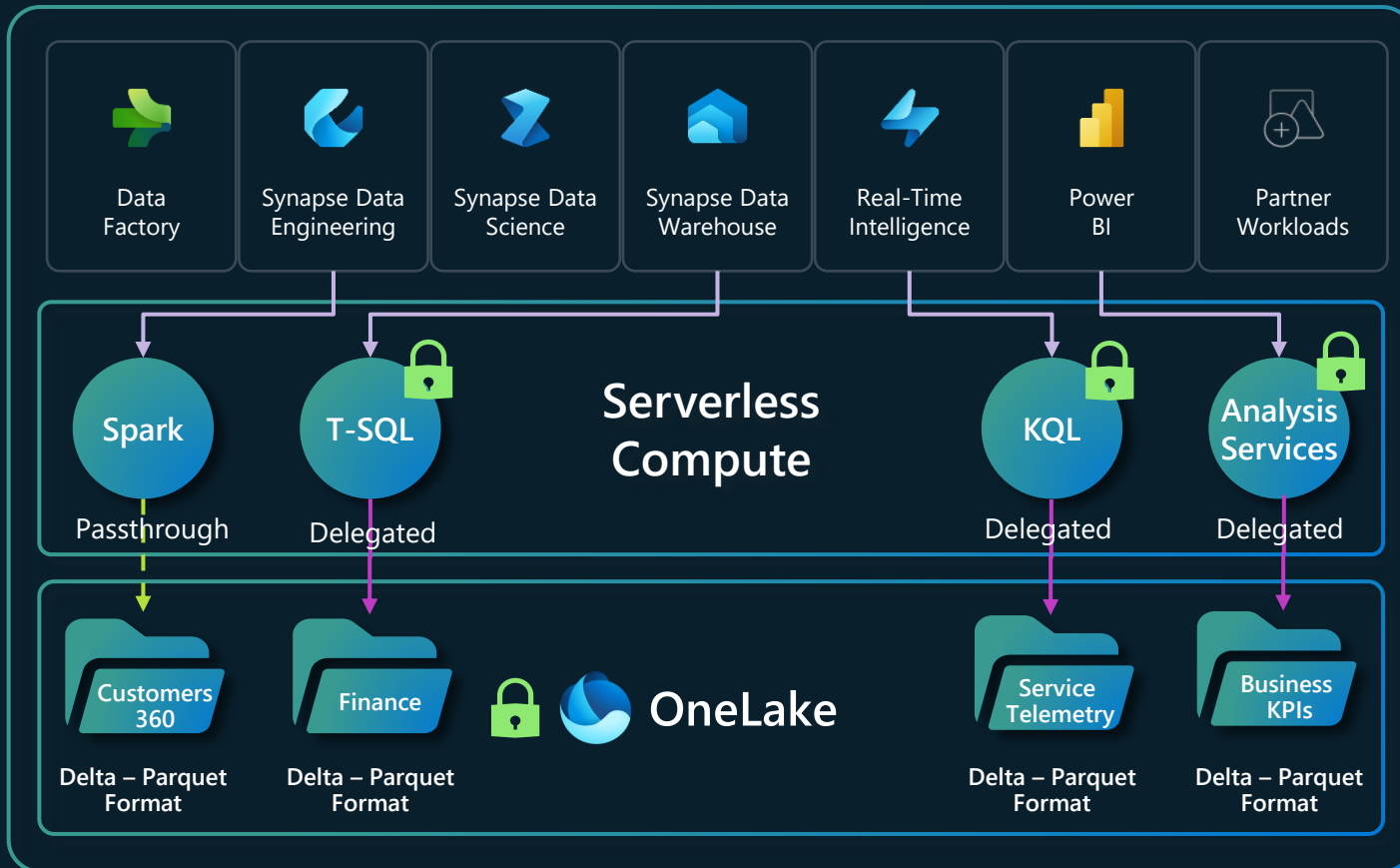
Delegated



In delegated mode, the trusted engine uses a fixed identity to check the access. UserA will then have access to the tables UserB has access to. However, additional security can be set in the trusted engine to further limit access.

One Copy, many security definitions

Enabling fine-grained security at each engine



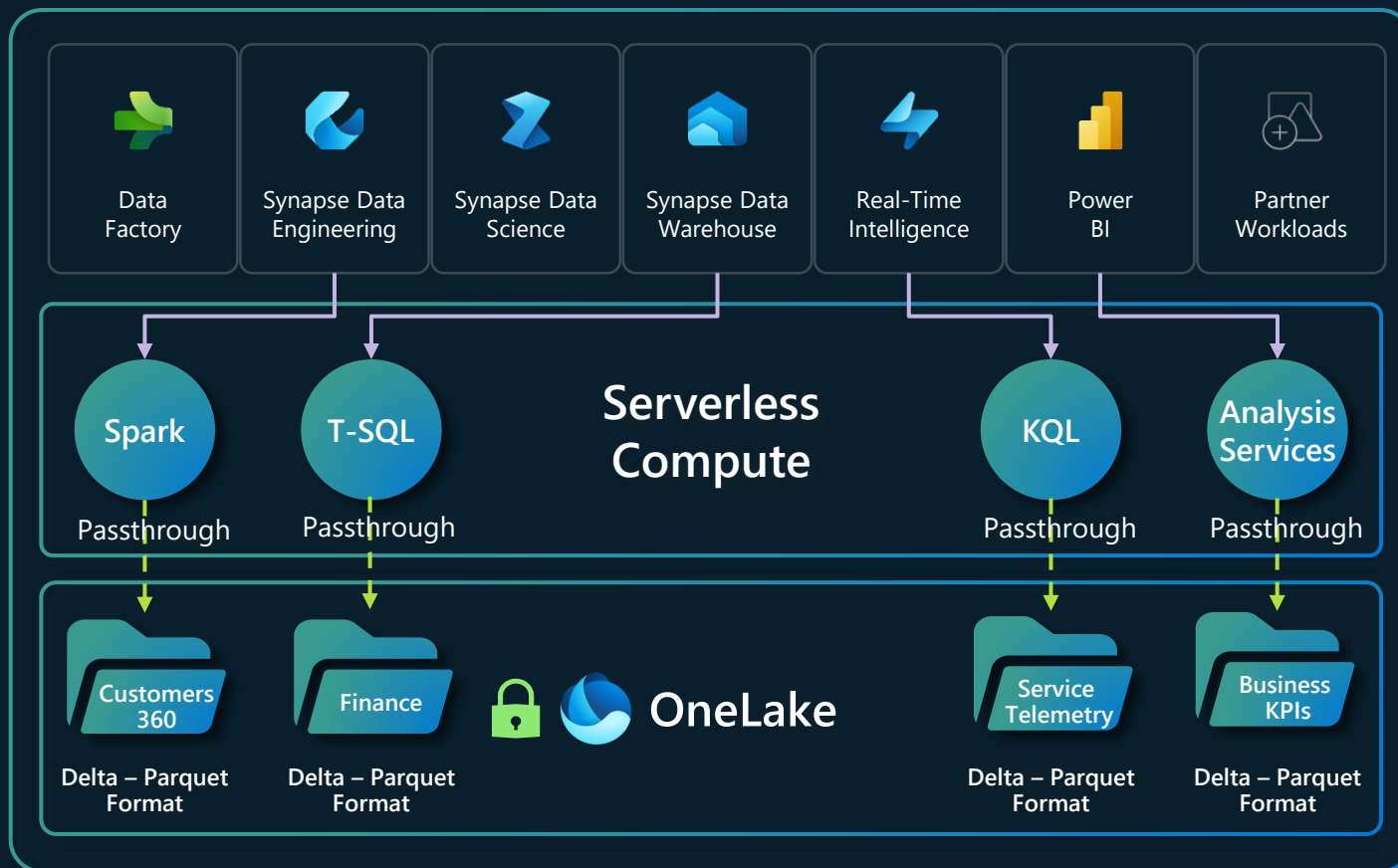
Most engines in Fabric use **delegated** mode to access the data.

Spark does not have native security, so it relies on **passthrough** mode to secure data.

In this state, customers have maximum flexibility of security. However, synchronizing security requires manual effort.

One Copy, One Security Definition

A single place to define and manage security



OneLake now has a complete set of fine-grained security features:

Table/folder security
Column level security
Row level security
+ more coming soon...

Engines can then switch to using passthrough mode as the primary security model.

Security is now centrally defined and enforced consistently across all engines.

OneLake security

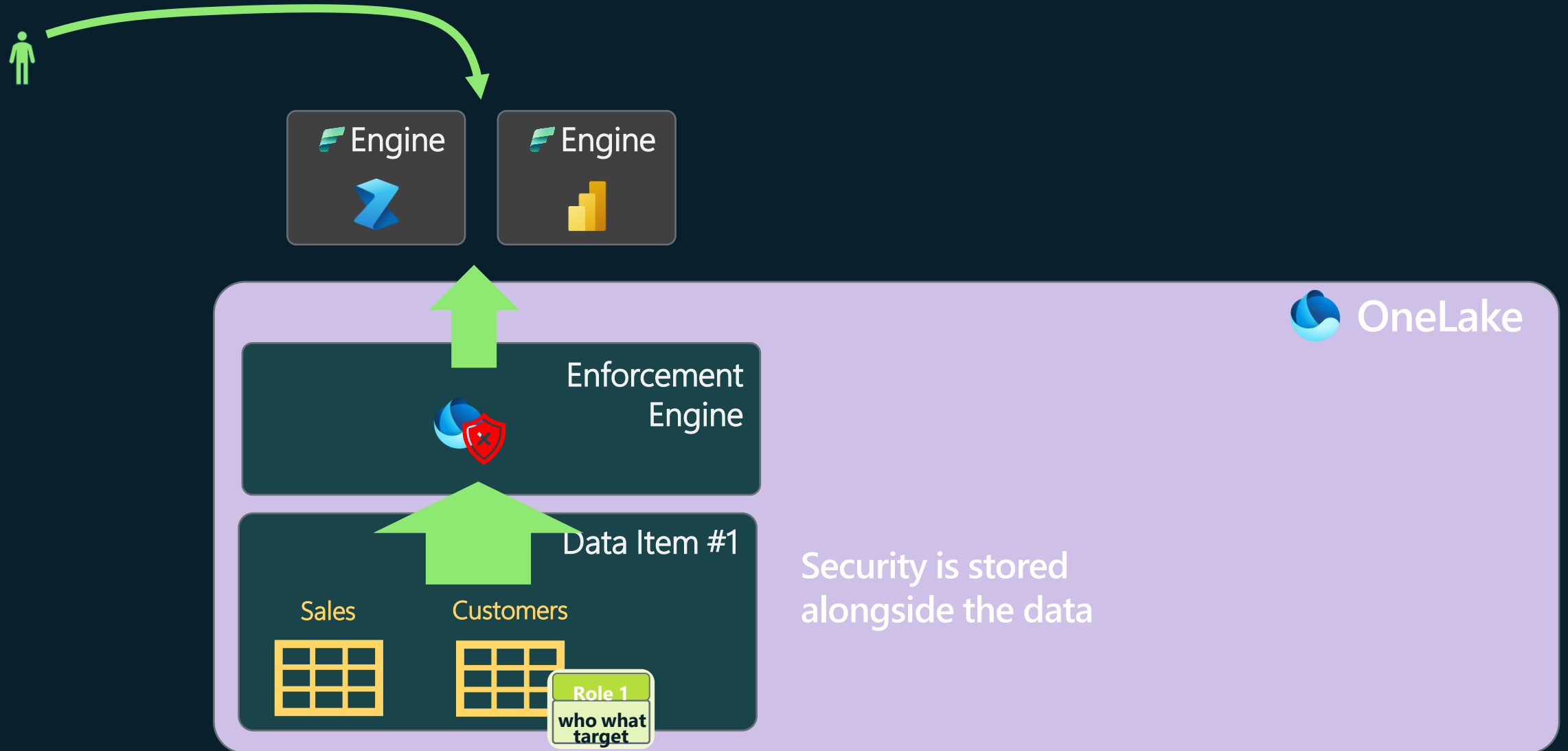
Principles

Defined once, enforced everywhere

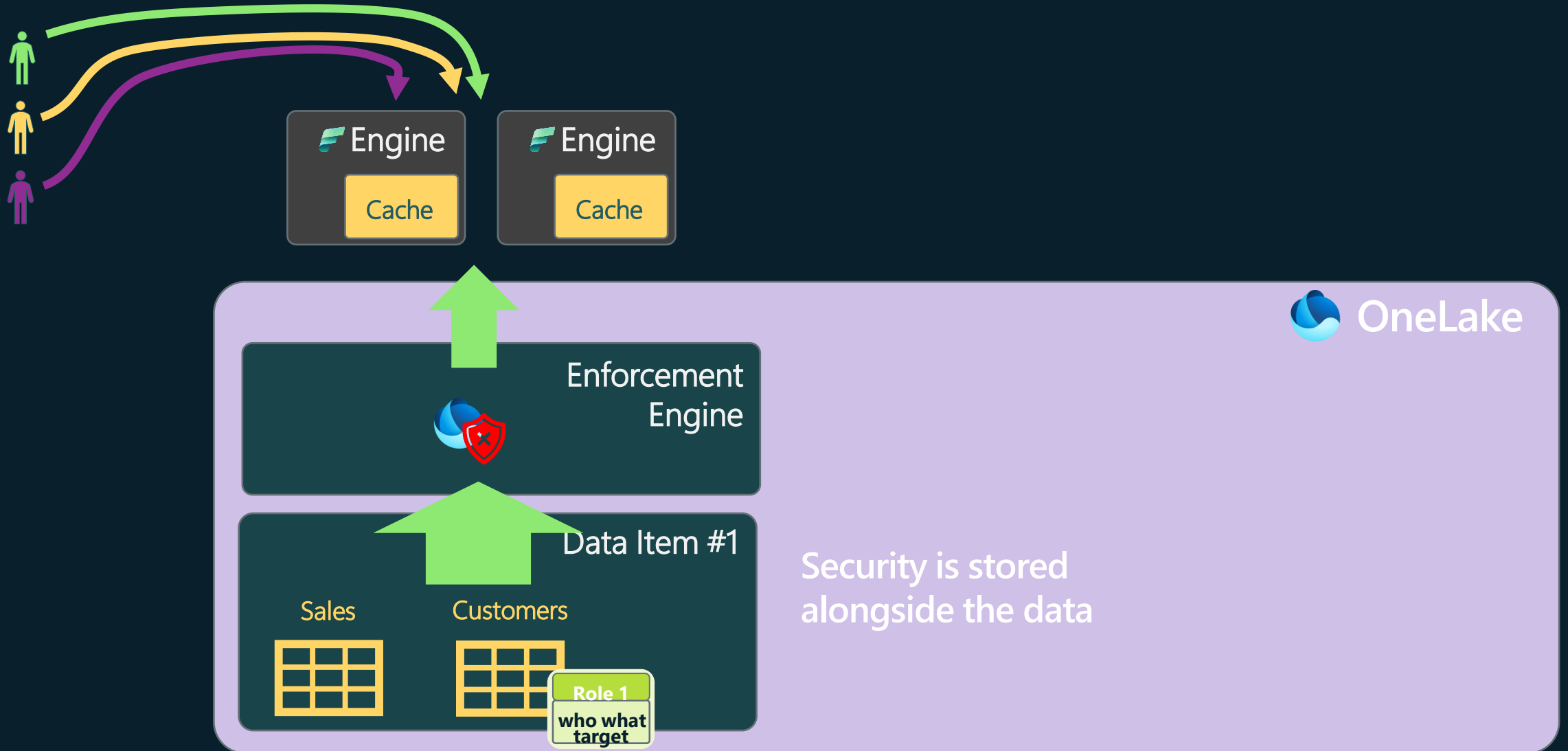
Security lives with the data

Enforce permissions safely

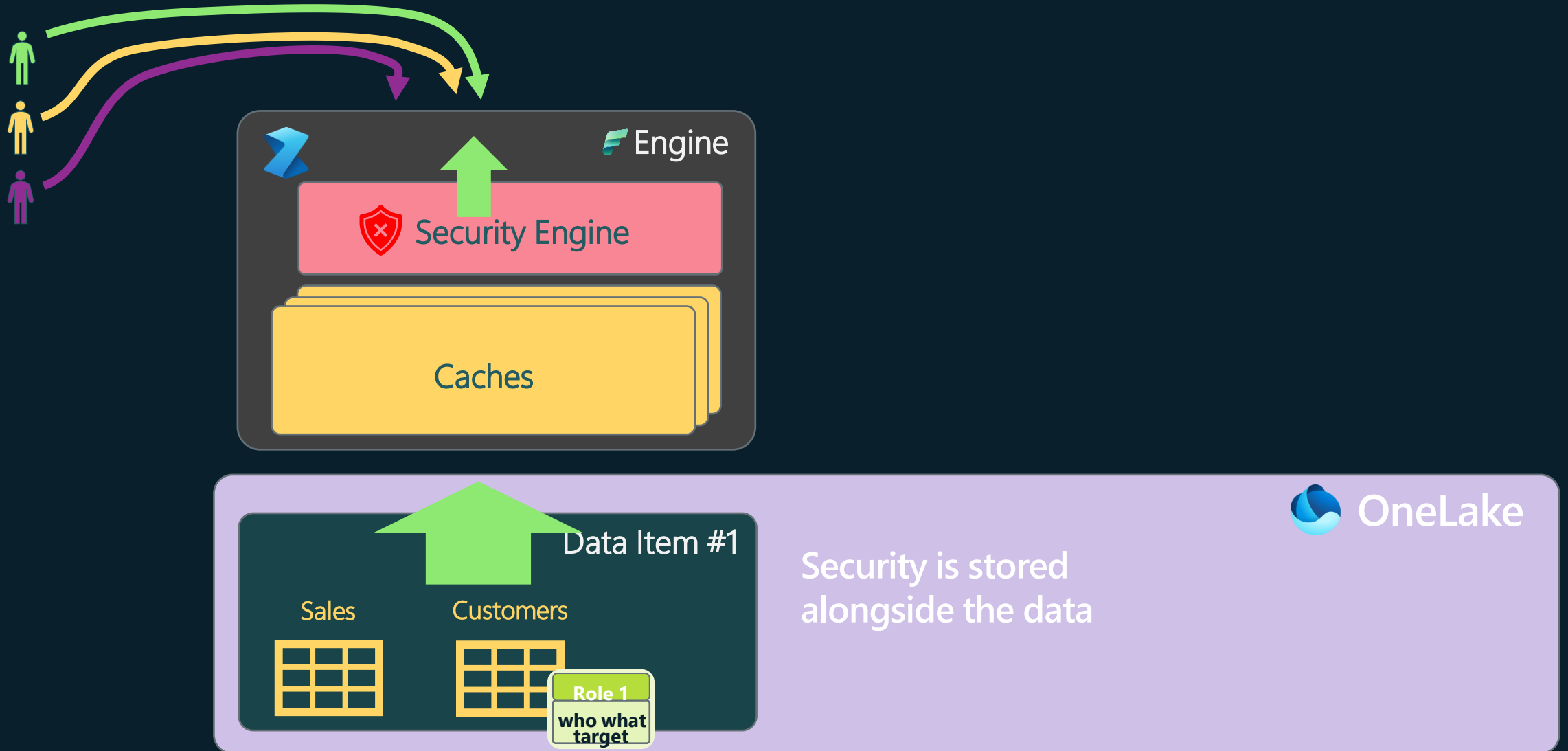
Enforcing OneLake security



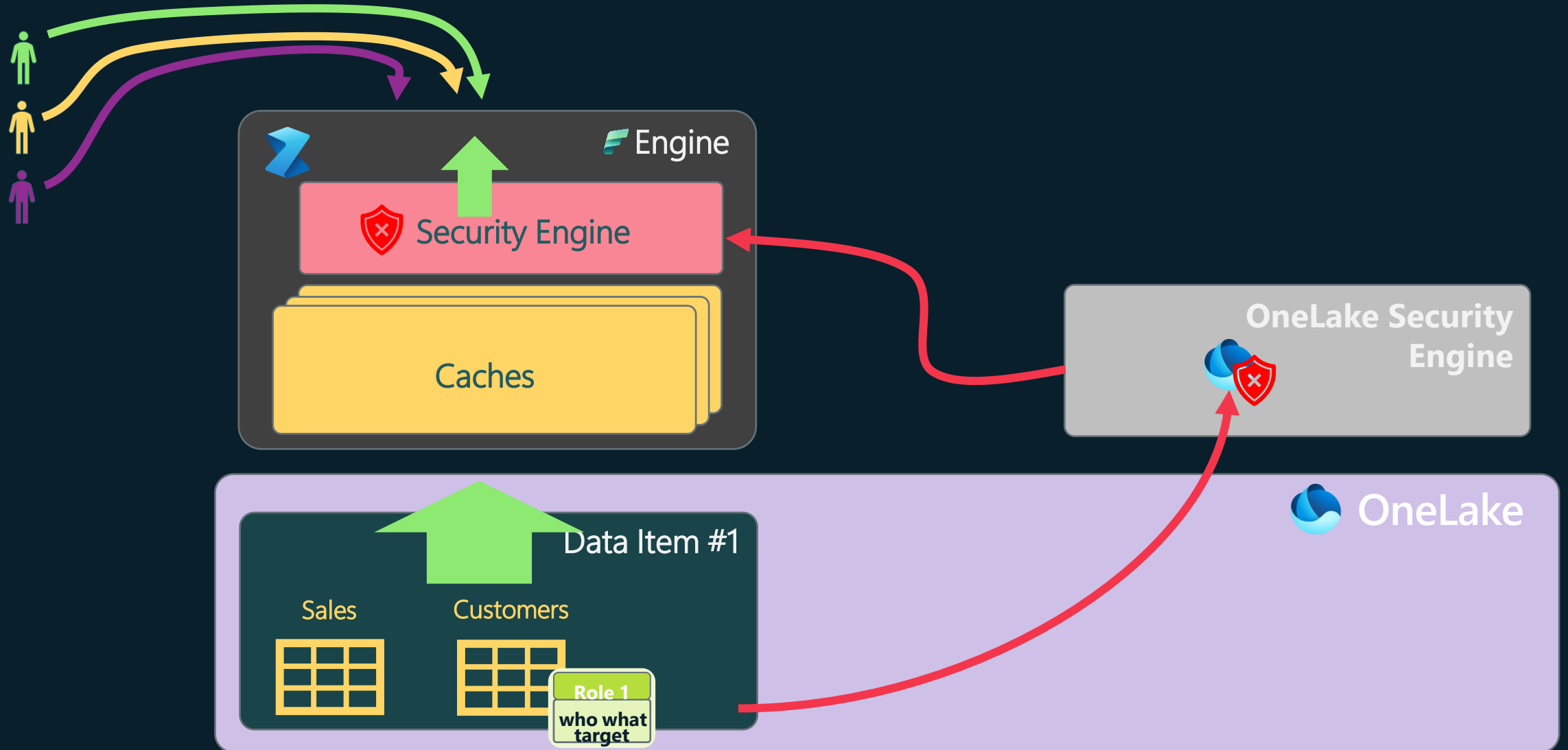
Enforcing OneLake security



Enforcing OneLake security



Enforcing OneLake security



Home

Get data ▾New semantic modelOpen notebook ▾Manage OneLake data access (preview)

Explorer

Search schemas and tables

NorthwindTest

Tables

dbo

Customers

Employees

EmployeeTerritor...

OrderDetails

Orders

Products

Regions

Shippers

ShortcutOrders

Suppliers

Territories

Files

Orders

Showing 830 rows

	123	orderID	ABC	customerID	123	employeeID	📅	orderDate	📅	requiredDate	ABC	shippedDate	123	shipVia	12	freight	ABC	shipName	ABC	shipAddress	ABC	shipCity	ABC	shipRegion	ABC	shipPostalC...
1		10257		HILAA		4		7/16/1996 12:00:...		8/13/1996 12:00:...		1996-07-22 00:0...		3		81.91		HILARION-Abast...		Carrera 22 con A...		San Cristóbal		Táchira		5022
2		10262		RATTC		8		7/22/1996 12:00:...		8/19/1996 12:00:...		1996-07-25 00:0...		3		48.29		Rattlesnake Can...		2817 Milton Dr.		Albuquerque		NM		87110
3		10268		GROSR		8		7/30/1996 12:00:...		8/27/1996 12:00:...		1996-08-02 00:0...		3		66.29		GROSELLA-Rest...		5ª Ave. Los Palos...		Caracas		DF		1081
4		10269		WHITC		5		7/31/1996 12:00:...		8/14/1996 12:00:...		1996-08-09 00:0...		1		4.56		White Clover Ma...		1029 - 12th Ave...		Seattle		WA		98124
5		10283		LILAS		3		8/16/1996 12:00:...		9/13/1996 12:00:...		1996-08-23 00:0...		3		84.81		LILA-Supermerc...		Carrera 52 con A...		Barquisimeto		Lara		3508
6		10287		RICAR		8		8/22/1996 12:00:...		9/19/1996 12:00:...		1996-08-28 00:0...		3		12.76		Ricardo Adocica...		Av. Copacabana ...		Rio de Janeiro		RJ		02389-890
7		10290		COMMI		8		8/27/1996 12:00:...		9/24/1996 12:00:...		1996-09-03 00:0...		1		79.7		Comércio Mineiro		Av. dos Lusíadas...		Sao Paulo		SP		05432-043
8		10296		LILAS		6		9/3/1996 12:00:0...		10/1/1996 12:00:...		1996-09-11 00:0...		1		0.12		LILA-Supermerc...		Carrera 52 con A...		Barquisimeto		Lara		3508
9		10305		OLDWO		8		9/13/1996 12:00:...		10/11/1996 12:0...		1996-10-09 00:0...		3		257.62		Old World Delic...		2743 Bering St.		Anchorage		AK		99508
10		10309		HUNGO		3		9/19/1996 12:00:...		10/17/1996 12:0...		1996-10-23 00:0...		1		47.3		Hungry Owl All-...		8 Johnstown Road		Cork		Co. Cork		NULL
11		10316		RATTC		1		9/27/1996 12:00:...		10/25/1996 12:0...		1996-10-08 00:0...		3		150.15		Rattlesnake Can...		2817 Milton Dr.		Albuquerque		NM		87110
12		10317		LONEP		6		9/30/1996 12:00:...		10/28/1996 12:0...		1996-10-10 00:0...		1		12.69		Lonesome Pine ...		89 Chiaroscuro ...		Portland		OR		97219
13		10324		SAVEA		9		10/8/1996 12:00:...		11/5/1996 12:00:...		1996-10-10 00:0...		1		214.27		Save-a-lot Mark...		187 Suffolk Ln.		Boise		ID		83720
14		10330		LILAS		3		10/16/1996 12:0...		11/13/1996 12:0...		1996-10-28 00:0...		1		12.75		LILA-Supermerc...		Carrera 52 con A...		Barquisimeto		Lara		3508
15		10338		OLDWO		4		10/25/1996 12:0...		11/22/1996 12:0...		1996-10-29 00:0...		3		84.21		Old World Delic...		2743 Bering St.		Anchorage		AK		99508
16		10346		RATTC		3		11/5/1996 12:00:...		12/17/1996 12:0...		1996-11-08 00:0...		3		142.08		Rattlesnake Can...		2817 Milton Dr.		Albuquerque		NM		87110
17		10347		FAMIA		4		11/6/1996 12:00:...		12/4/1996 12:00:...		1996-11-08 00:0...		3		3.1		Familia Arquibal...		Rua Orós 92		Sao Paulo		SP		05442-030
18		10349		SPLIR		7		11/8/1996 12:00:...		12/6/1996 12:00:...		1996-11-15 00:0...		1		8.63		Split Rail Beer & ...		P.O. Box 555		Lander		WY		82520
19		10355		AROUT		6		11/15/1996 12:0...		12/13/1996 12:0...		1996-11-20 00:0...		1		41.95		Around the Horn		Brook Farm Strat...		Colchester		Essex		CO7 6JX
20		10357		LILAS		1		11/19/1996 12:0...		12/17/1996 12:0...		1996-12-02 00:0...		3		34.88		LILA-Supermerc...		Carrera 52 con A...		Barquisimeto		Lara		3508
21		10373		HUNGO		4		12/5/1996 12:00:...		1/2/1997 12:00:0...		1996-12-11 00:0...		3		124.12		Hungry Owl All-...		8 Johnstown Road		Cork		Co. Cork		NULL
22		10379		QUEDE		2		12/11/1996 12:0...		1/8/1997 12:00:0...		1996-12-13 00:0...		1		45.03		Que Delícia		Rua da Panificad...		Rio de Janeiro		RJ		02389-673
23		10380		HUNGO		8		12/12/1996 12:0...		1/9/1997 12:00:0...		1997-01-16 00:0...		3		35.03		Hungry Owl All-...		8 Johnstown Road		Cork		Co. Cork		NULL
24		10381		LILAS		3		12/12/1996 12:0...		1/9/1997 12:00:0...		1996-12-13 00:0...		3		7.99		LILA-Supermerc...		Carrera 52 con A...		Barquisimeto		Lara		3508
25		10383		AROUT		8		12/16/1996 12:0...		1/13/1997 12:00:...		1996-12-18 00:0...		3		34.24		Around the Horn		Brook Farm Strat...		Colchester		Essex		CO7 6JX
26		10386		FAMIA		9		12/18/1996 12:0...		1/1/1997 12:00:0...		1996-12-25 00:0...		3		13.99		Familia Arquibal...		Rua Orós 92		Sao Paulo		SP		05442-030
27		10393		SAVEA		1		12/25/1996 12:0...		1/22/1997 12:00:...		1997-01-03 00:0...		3		126.56		Save-a-lot Mark...		187 Suffolk Ln.		Boise		ID		83720
28		10394		HUNGC		1		12/25/1996 12:0...		1/22/1997 12:00:...		1997-01-03 00:0...		3		30.34		Hungry Coyote l...		City Center Plaza...		Elgin		OR		97827
29		10395		HILAA		6		12/26/1996 12:0...		1/23/1997 12:00:...		1997-01-03 00:0...		1		184.41		HILARION-Abast...		Carrera 22 con A...		San Cristóbal		Táchira		5022
30		10398		SAVEA		2		12/30/1996 12:0...		1/27/1997 12:00:...		1997-01-09 00:0...		3		89.16		Save-a-lot Mark...		187 Suffolk Ln.		Boise		ID		83720
31		10401		RATTC		1		1/1/1997 12:00:0...		1/29/1997 12:00:...		1997-01-10 00:0...		1		12.51		Rattlesnake Can...		2817 Milton Dr.		Albuquerque		NM		87110
32		10405		LINOD		1		1/6/1997 12:00:0...		2/3/1997 12:00:0...		1997-01-22 00:0...		1		34.82		LINO-Delicateses		Ave. 5 de Mayo ...		I. de Margarita		Nueva Esparta		4980
33		10406		QUEEN		7		1/7/1997 12:00:0...		2/18/1997 12:00:...		1997-01-13 00:0...		1		108.04		Queen Cozinha		Alameda dos Ca...		Sao Paulo		SP		05487-020
34		10410		BOTTM		3		1/10/1997 12:00:...		2/7/1997 12:00:0...		1997-01-15 00:0...		3		2.4		Bottom-Dollar ...		23 Tsawassen Bl...		Tsawassen		BC		T2F 8M4

Succeeded (7 sec 28 ms)

Columns 14 Rows 830

OneLake security - details

Concepts and overview

OneLake security uses a role based approach to define access to data. Users are granted permissions to data based on the roles they are members of.

Roles

The core concept of permissions management in OneLake. Roles represent a set of permissions, applied to a scope, for a given set of members.

Permissions

A set of actions that users are allowed to take on a specified scope. Permissions are things like Read, Write, Create, etc.

Scope

A scope is the set of items you are granting permissions to. It can be a table, folders, or an artifact. It can also have constraints that restrict access to specific rows or columns.

Members

Members are the users or identities that are accessing data. Members are given the permissions of a role by being assigned to that role. Members can be groups or non-user identities such as service principals.

OneLake security types

OneLake security supports three primary types of security with additional features planned for future iterations.

Folder

The base security type in OneLake, define permissions for a Folder in OneLake. Permissions inherit to subfolders for easy management.

Folders in the Tables path of a Fabric artifact contain tabular data and can be assigned additional permissions.

Row

Limit access to specific rows of data in a table. Users specify SQL predicates that are used to restrict access.

OR

Column

Restrict access to data in certain columns using column level security. Define specific permissions on individual columns to hide sensitive data. Data masking is also an option as part of column level security.

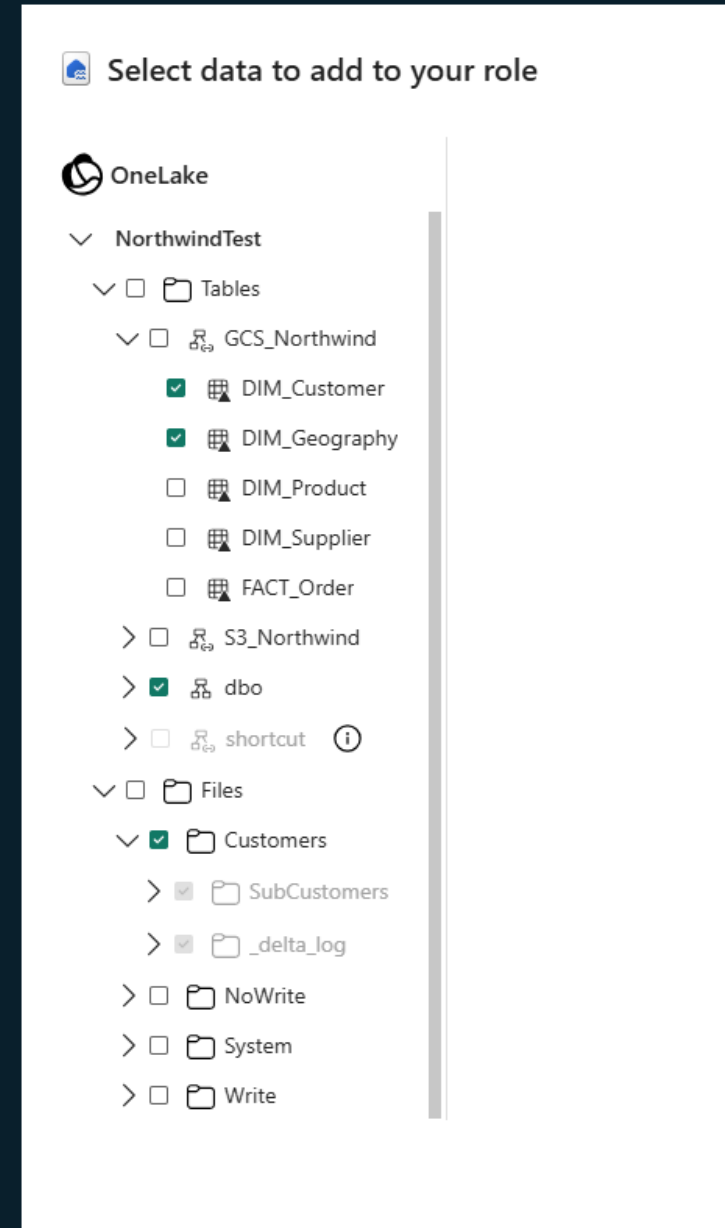
OneLake – object level security

Folder level security allows you to set permissions on a folder or folders in OneLake.

Security is based on Windows and SharePoint access models, as opposed to POSIX style ACLs.

Permissions inherit automatically to any sub folders.

Tables are a subset of folder security that can also have more granular permission types applied to them.



OneLake – column level security

Column level security allows for limiting access to specific columns in a table.

Access to columns can be left to default or columns can be removed, preventing users from seeing the column.

Hidden columns will be removed automatically when querying in Spark and Power BI. SQL requires excluding the columns from a query.

Data masking is not available yet.

Manage table level access for Customers

Manage access to your data by adding row and column level security to a table in a role. Members of this role will only be able to see specific data based on the rules you set below. [Learn more](#)

Row security (Preview)

Column security (Preview)

Remove

Column level security

Enabled

Column

Visibility

☒	companyName	Read
☒	contactName	Read
	contactTitle	Read
	address	Read
	city	Read
	region	Read
	postalCode	Read

OneLake – row level security

Row level security allows for defining security predicates to govern access to select rows within a table.

Predicates use T-SQL syntax for definition and allow for multiple conditions to be set.

RLS can be dynamic per user or lookup data across multiple tables. (coming soon)

Manage table level access for

Customers

▼

Manage access to your data by adding row and column level security to a table in a role. Members of this role will only be able to see specific data based on the rules you set below. [Learn more](#)

Row security (Preview)

Column security (Preview)

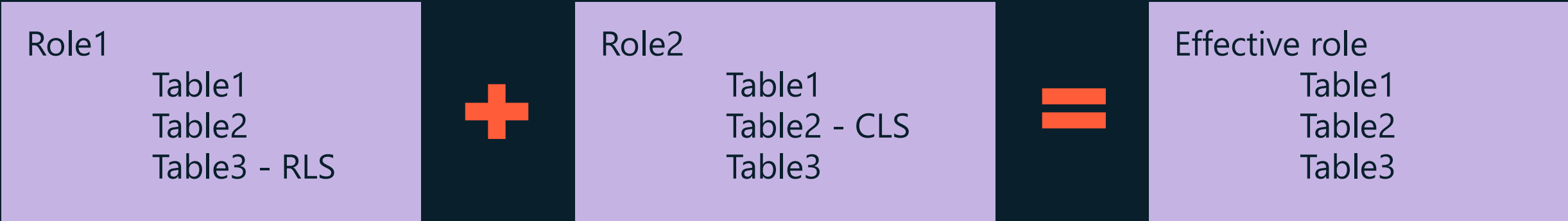
Save

Show data to members of your role if the following rules apply:

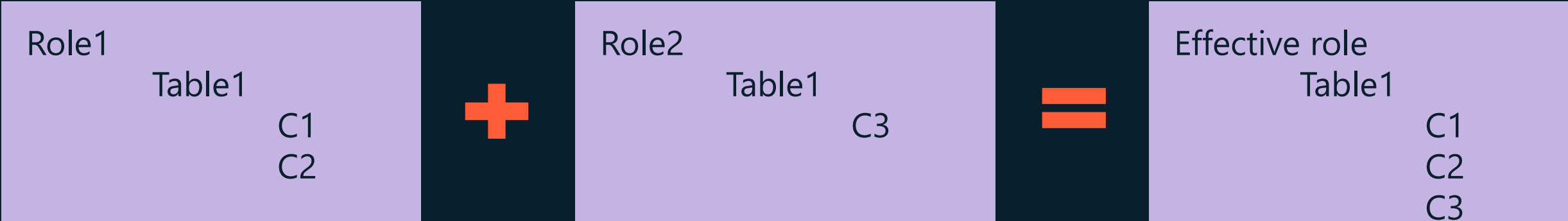
```
1 SELECT * FROM dbo.Customers WHERE country = 'USA'
```

OneLake – role interaction

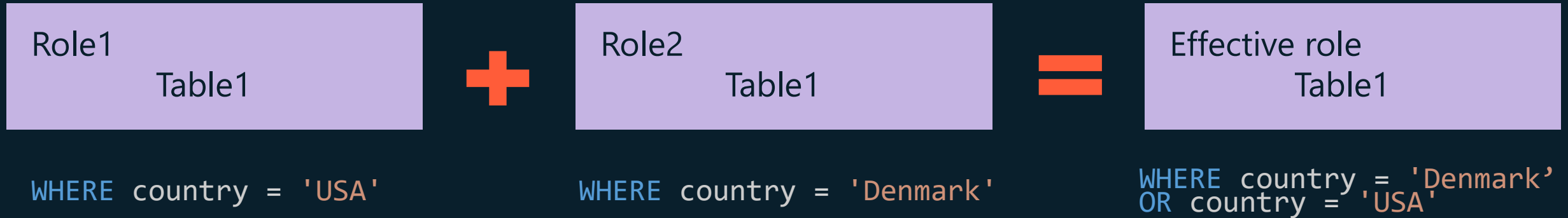
OneLake security roles
combine using **union**
behavior (most permissive)



OneLake – role interaction

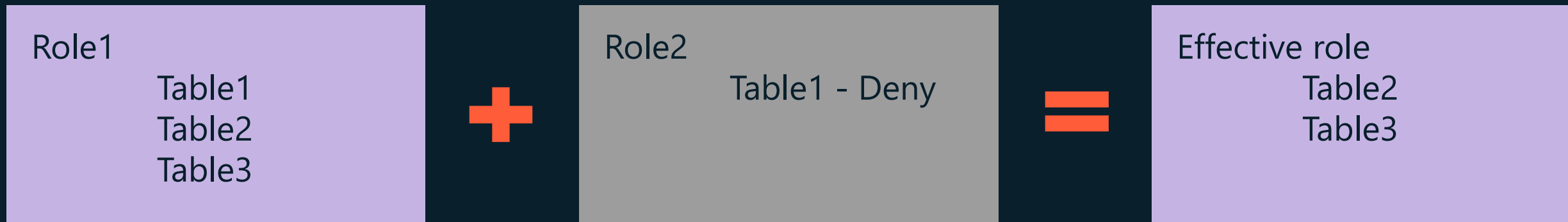


OneLake – role interaction



OneLake – role interaction

Deny roles always override
any other role



OneLake security – API details

OneLake security - API

Interacting with OneLake security programmatically

GET/PUT dataAccessRoles

GET/PUT

<https://api.fabric.microsoft.com/v1/workspaces/{workspaceId}/items/{itemId}/dataAccessRoles>

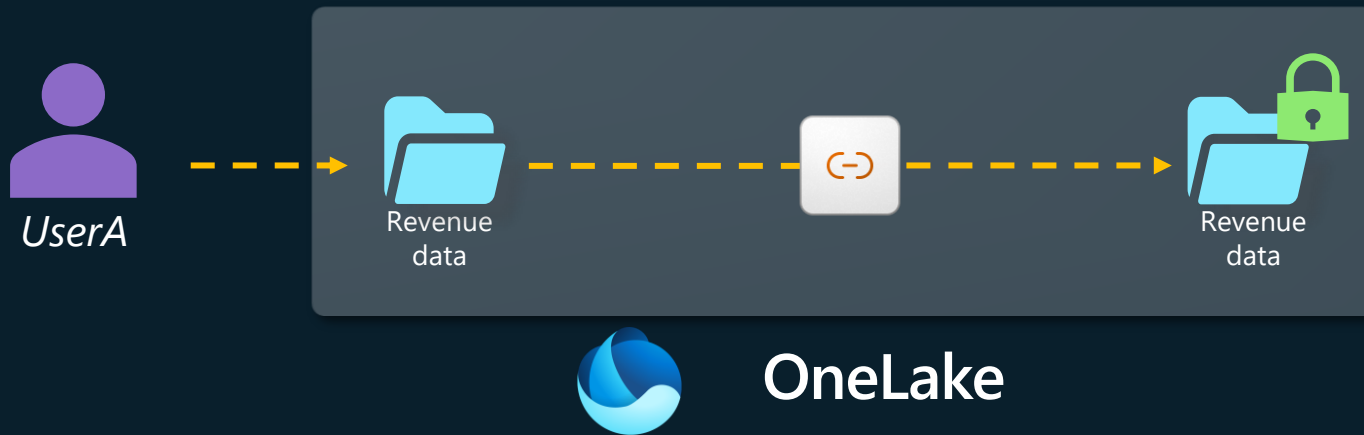
This is a bulk API. The existing role payload in the item gets replaced by the new one being sent.

Shortcuts and integration patterns

OneLake security - shortcuts

Shortcuts are either passthrough or delegated

Passthrough



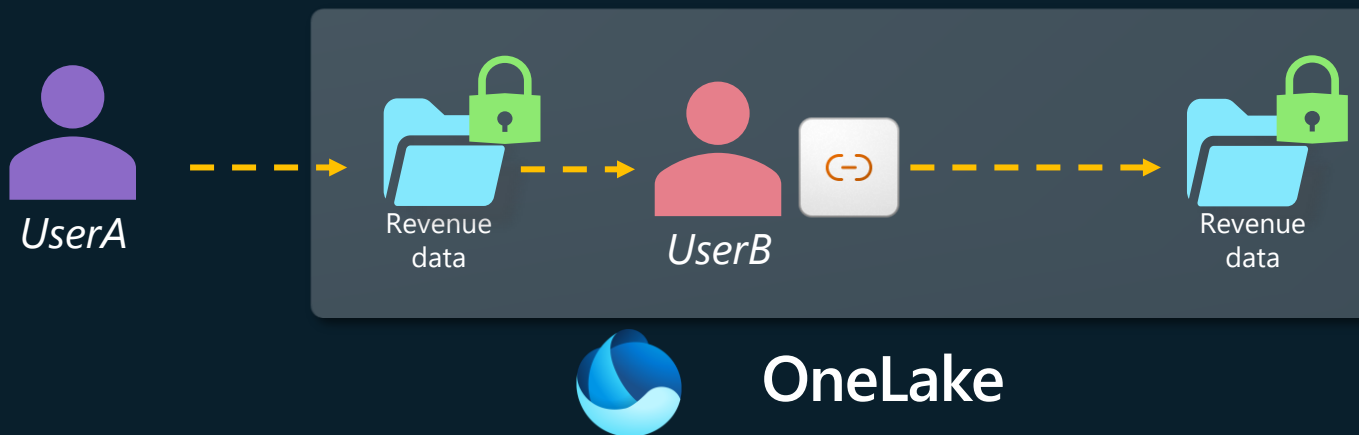
Shortcuts in OneLake that are passthrough:

- OneLake shortcuts

OneLake security - shortcuts

Shortcuts are either passthrough or delegated

Delegated

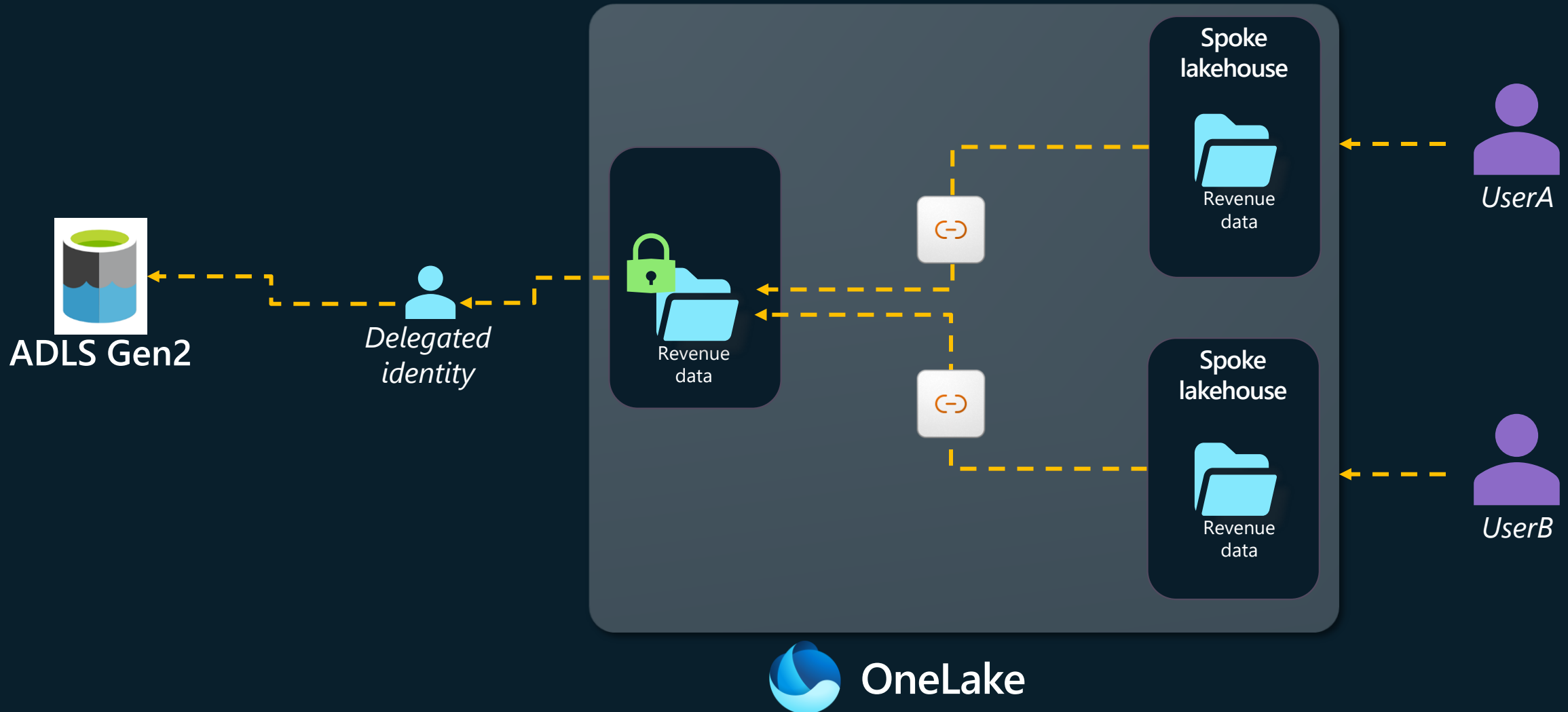


Shortcuts in OneLake that are delegated:

- AWS S3
- GCS
- ADLS Gen2
- etc

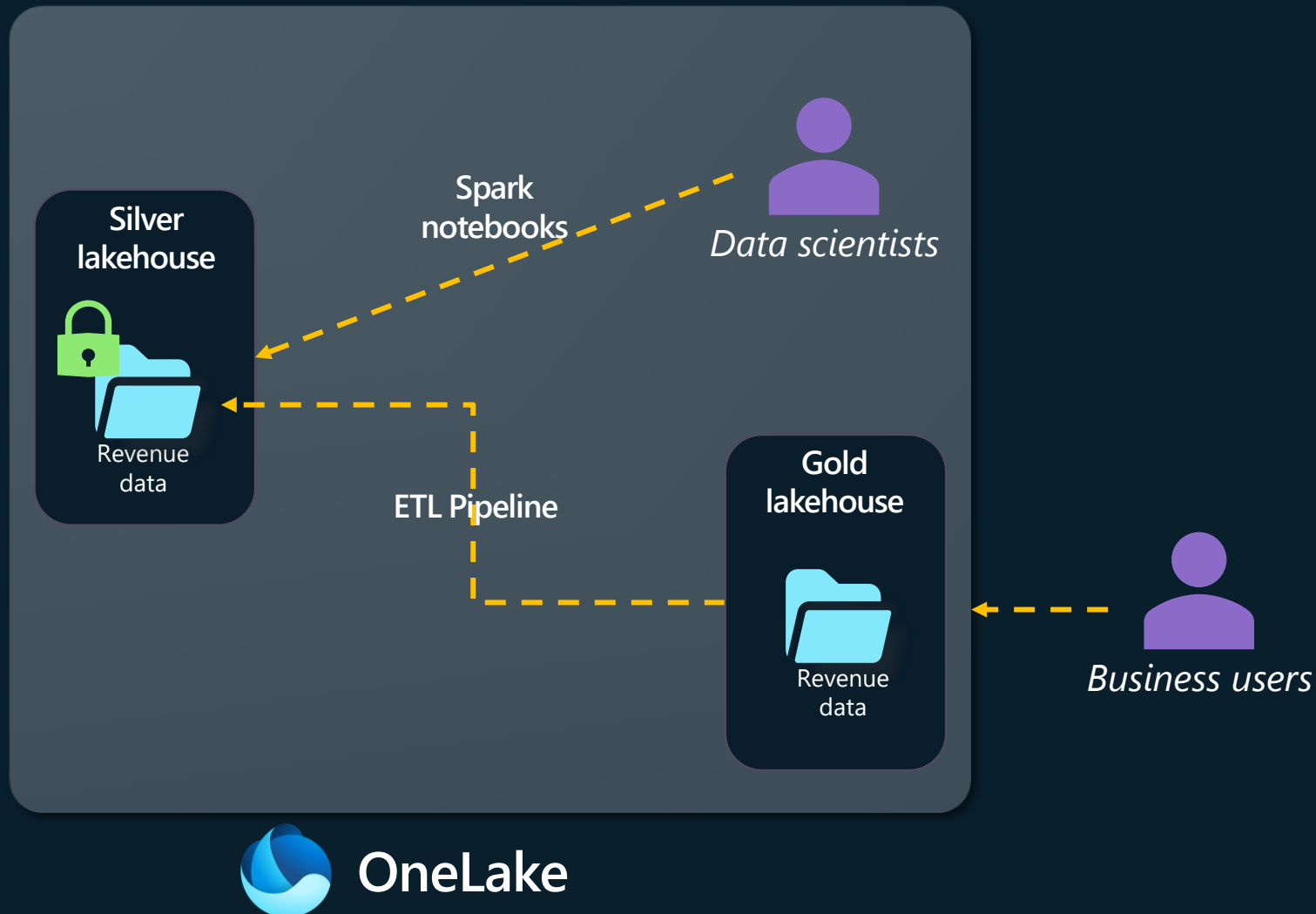
OneLake security - patterns

Shortcuts as a distribution method



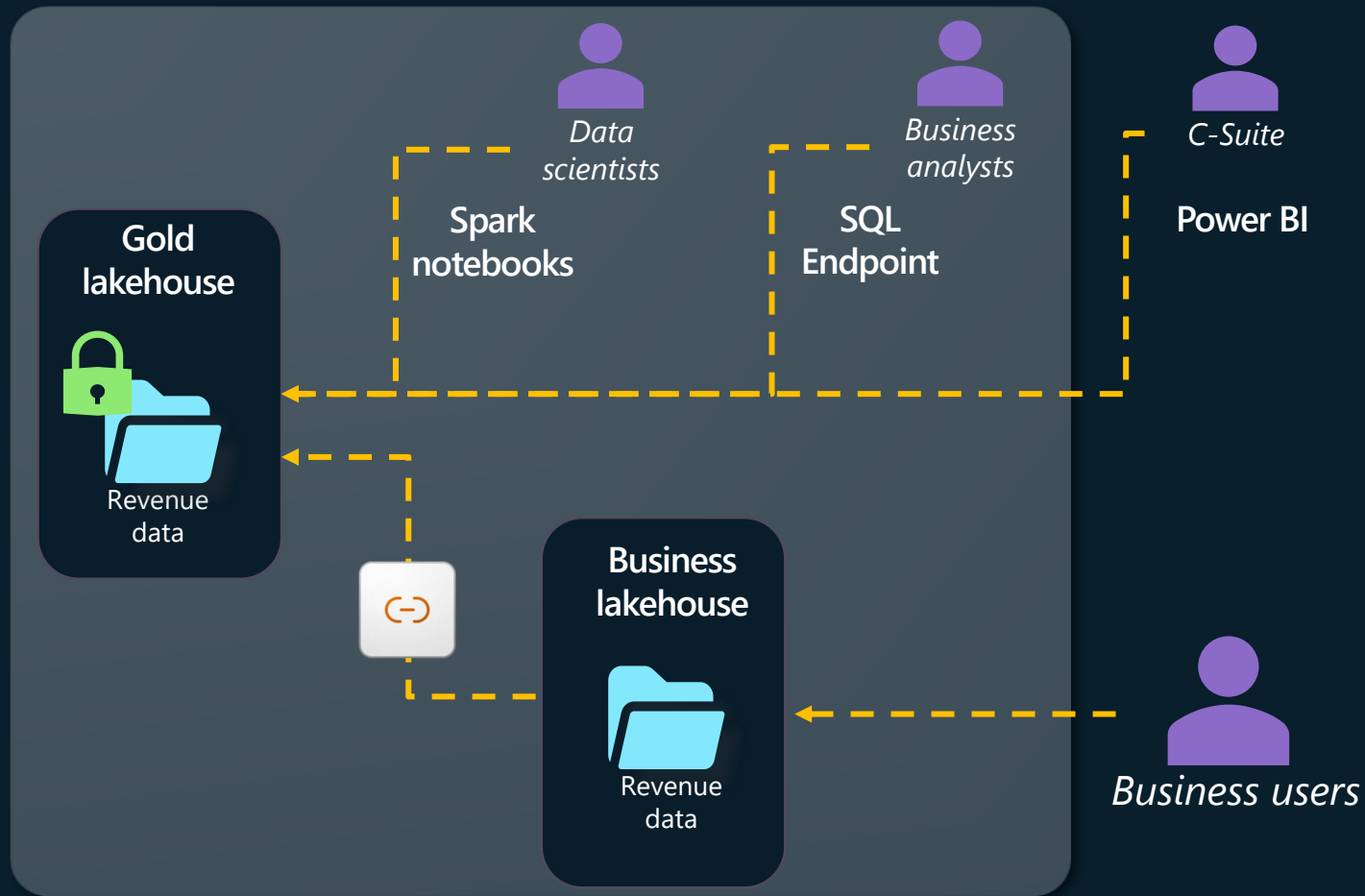
OneLake security - patterns

OneLake security for a silver layer



OneLake security - patterns

OneLake security for a gold layer



Managing OneLake security

Home

Create

Browse

OneLake

Monitor

Deployment pipelines

Real-Time hub

Functions hub

Workspaces

My workspace

Synapse Data Engineering OneLake

Search

Trial: 18 days left

OneLake catalog

Domain: All domains

Explore

Govern

Secure

View users

View security roles

Workspaces

My workspace

Oxnard

Rock Hill

St. Paul

South Jordan

More workspaces...

Select data to add to your role

Lakehouse Name

Schemas

Schema1

Tables

Table

Schema2

Tables

Files

Tree item

Tree item

Tree item

Tree item

Table1 Name

Schemas > Schema1 > Tables > Table1

Name	File Type	Last modified
Content	Folder	11/23/18, 1:00 PM
Content	File	11/23/18, 1:00 PM
Content	File	11/23/18, 1:00 PM
Content	File	11/23/18, 1:00 PM
Content	File	11/23/18, 1:00 PM
Content	File	11/23/18, 1:00 PM
Content	File	11/23/18, 1:00 PM
Content	File	11/23/18, 1:00 PM
Content	File	11/23/18, 1:00 PM
Content	File	11/23/18, 1:00 PM
Content	File	11/23/18, 1:00 PM
Content	File	11/23/18, 1:00 PM

Add data

DefaultRole

Product Metrics

DefaultRole

Product Metrics

DefaultRole

Product Metrics

Schema A

Schema

Read

A

Table

Read

B

Table

Read

Home

Create

Browse

OneLake

Monitor

Deployment pipelines

Real-Time hub

Functions hub

Workspaces

My workspace

Data Engineering

OneLake catalog

All domains

Explore

Govern

Secure

- All items
- My items
- Endorsed in your...
- Favorites
- Workspaces
- Rock Hill
- Oxnard
- South Jordan
- St. Paul
- South
- Contoso workspace
- More workspaces...

1 Selected | Refresh | Favorite | Add tag | ...

Details

Name

> Sales FY21

> Customer feedback

> Sales FY21

> Test datamart

> Primary dataflow

> Primary dataflow

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

Data Warehouse

Customer feedback

Open

Overview | Lineage | Permissions | Query activity

Location

Contoso > Folder name

Data updated

1/26/25, 3:02 PM

Contact

Dani Wolf

Sensitivity label

Confidential

SQL connection string

x6eps4xrq2xudenlfv6naeo3...

+ 2

Objects (17)

Name	Date modified	Type	Size
> queryinsights	10/18/2023 at 11:56 AM	Table	470 B
> dbo	10/03/2023 at 09:23 AM	Table	1 G
> information_schema	10/03/2023 at 09:23 AM	Table	2 G
> _delta_log	10/03/2023 at 09:23 AM	Table	256 B
> check_constraints	10/03/2023 at 09:23 AM	View	-
> KEY_COLUMN_USAGE	10/03/2023 at 09:23 AM	View	-
> Usage schema	10/03/2023 at 09:23 AM	View	-
> _delta_log	10/03/2023 at 09:23 AM	Table	256 B
> _delta_log	10/03/2023 at 09:23 AM	Table	256 B

 Wanda Howard

OneLake catalog

All domains

Explore

Govern

Secure

- All items
- My items
- Endorsed in your...
- Favorites
- Workspaces
- Rock Hill
- Oxnard
- South Jordan
- St. Paul
- South
- Contoso workspace
- More workspaces...

1 Selected | Refresh | Favorite | Add tag | ...

Details

Name

> Sales FY21

> Customer feedback

> Sales FY21

> Test datamart

> Primary dataflow

> Primary dataflow

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

> Dataflow main

Data Warehouse

Customer feedback

This warehouse shows IT spend YoY. this is the default text for the item's description if defined.

Open

Overview | Lineage | Permissions

Direct access | External data shares | OneLake Security | SQL Security | Audiences (only PBI)

+ New

Role name	Permission
DefaultRole	Read
Sales USA	Read, SQL Permissions



Wrap up

OneLake security feature - now

Define OneLake security on:

- Lakehouse
- Azure Mirrored Databrick Catalog

OneLake security filtering:

- Lakehouse
- Spark notebooks
- SQL Analytics Endpoint
- Semantic Models

OneLake security – upcoming

Planned within the next 6 months

Open public preview

Dynamic and multi-table RLS

Strong deny

Write permission for folders

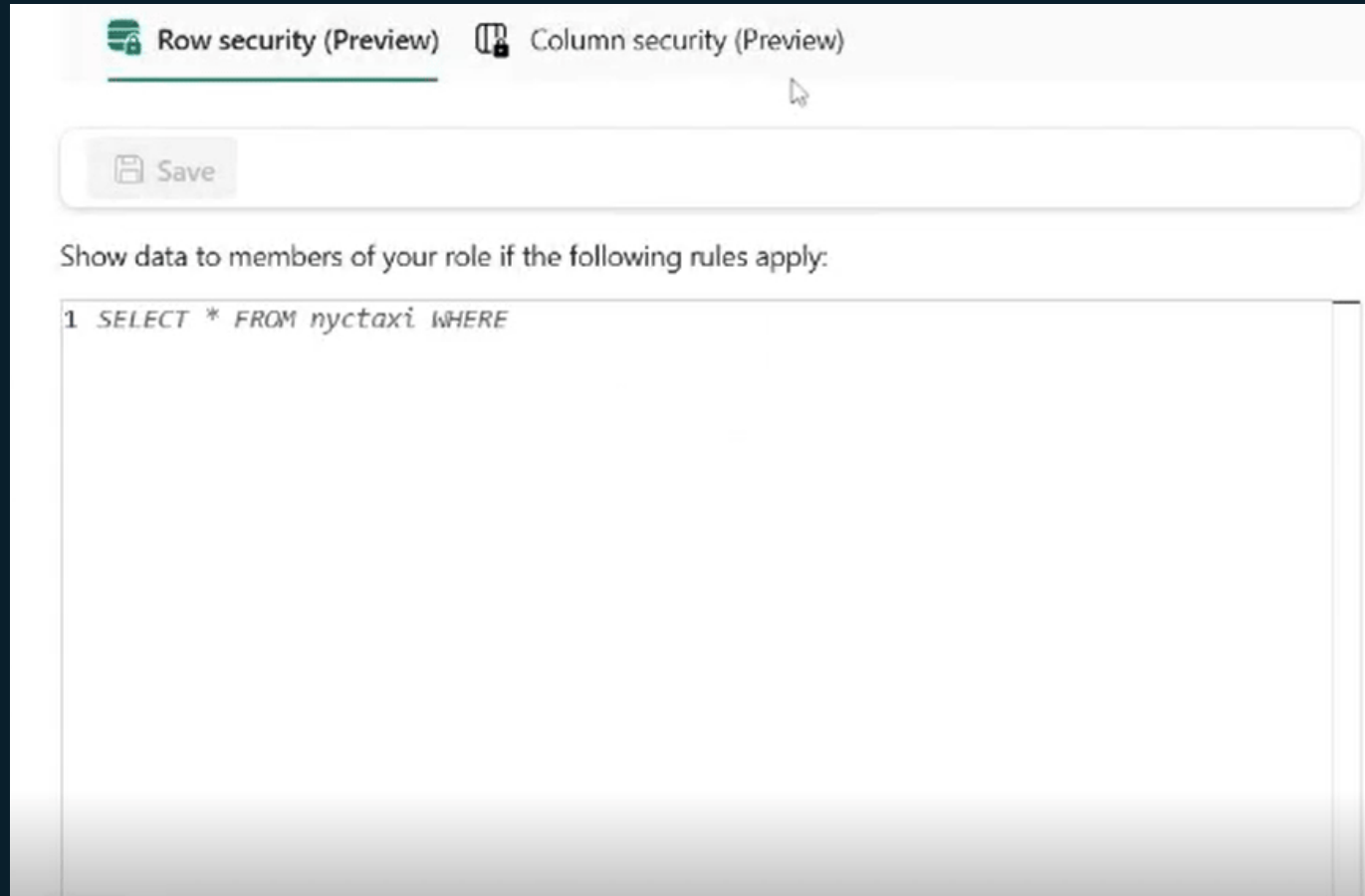
More granular public APIs

Additional items and engines:

Event House - shortcuts

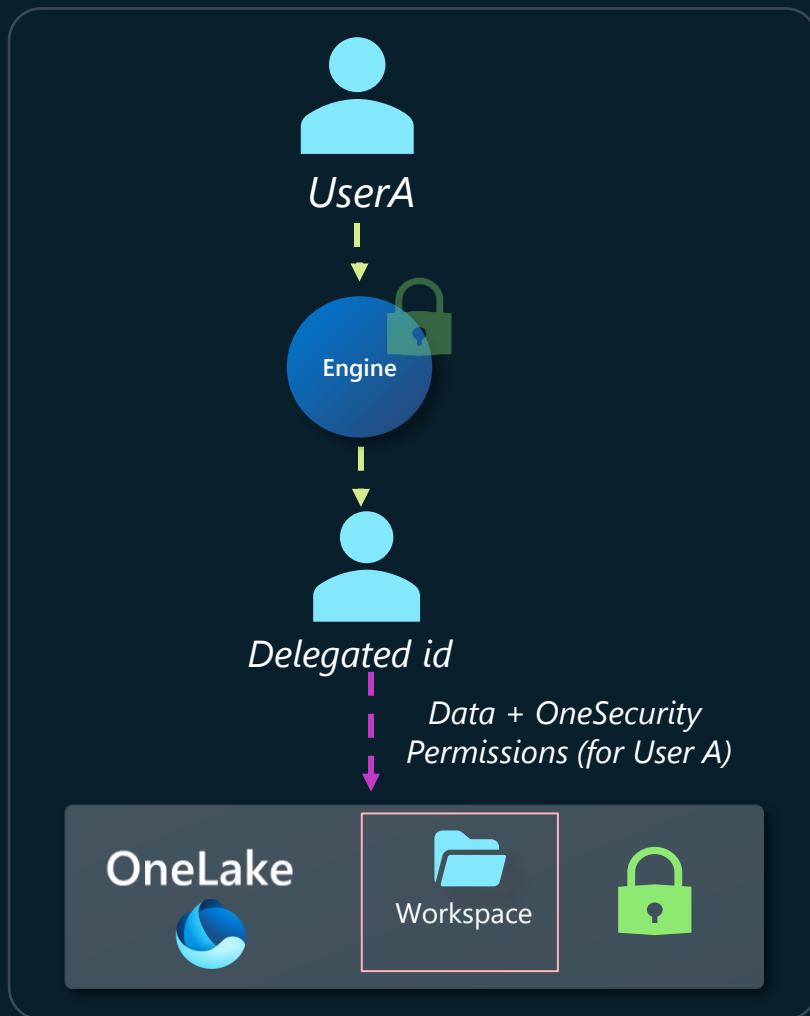
RLS Intellisense

Available in June



Open OneLake security

Enabling fine-grained security for *any* engine



Delegation w/ Impersonation

The engine uses a fixed (delegated) identity to access data **and** enforces calling user access as defined in OneLake.

Users decide which engines to trust for security enforcement, with the ability to revoke access at any time.

Requires work by the engine to support OneLake security through APIs.

Next steps

OneLake security - early access
signup form



Next steps

Currently in preview

Signups are open to anyone using the QR code or at:

aka.ms/OneLakeSecurityPreview

New users will be enabled each week in the order they are received.

Q&A

Session eval



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